

Question number	Answer	Additional guidance	Mark
4(a)(i)	The correct answer is D <i>A is incorrect because the vena cava carries blood to the heart, has valves along its length and has an endothelial lining</i> <i>B is incorrect because the vena cava carries blood to the heart, has valves along its length and has an endothelial lining</i> <i>C A is incorrect because the vena cava carries blood to the heart, has valves along its length and has an endothelial lining</i>		(1)

Question number	Answer	Additional guidance	Mark
4(a)(ii)	The correct answer is C <i>A is incorrect because the direction of flow of blood away from the heart classifies a blood vessel as an artery</i> <i>B is incorrect because the direction of flow of blood away from the heart classifies a blood vessel as an artery</i> <i>D is incorrect because arteries carry blood away from the heart not towards it.</i>		(1)

Question number	Answer	Additional guidance	Mark
4(a)(iii)	The correct answer is C <i>A is incorrect because the coronary artery branches from the aorta and P is the vena cava</i> <i>B is incorrect because the coronary artery branches from the aorta and Q is the pulmonary artery</i> <i>D is incorrect because the coronary artery branches from the aorta and S is the pulmonary vein</i>		(1)

Question number	Answer	Additional guidance	Mark
4(a)(iv)	The correct answer is A <i>B is incorrect because the valve T is the bicuspid valve and is open during both atrial systole and diastole</i> <i>C is incorrect because the valve T is the bicuspid valve and is open during both atrial systole and diastole</i> <i>D is incorrect because the valve T is the bicuspid valve and is open during both atrial systole and diastole</i>		(1)

Question number	Answer	Additional guidance	Mark
4(b)(i)	32 (beats min⁻¹)		(1)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	0.5 / 0.54 / 0.541 (seconds)		(1)

Question number	Answer	Additional guidance	Mark
4(b)(iii)	An answer that includes three of the following points: - atropine increases heart rate and propranolol decreases heart rate (1) - atropine increases heart rate whether given first or second (1) OR propranolol decrease heart rate whether given first or second (1) - either regimen results in {same / similar} (overall) heart rate (1) - credit suitable comment about error bars (1)	ACCEPT propranolol decreases heart rate more when added after atropine ACCEPT order of drugs makes {little / no difference} overall both groups ended up the same e.g. these results are significant as error bars do not overlap (except in last two days) error bars are close together so drugs are having similar effects error bars for atropine (in first part) are wider so more uncertainty	(3)